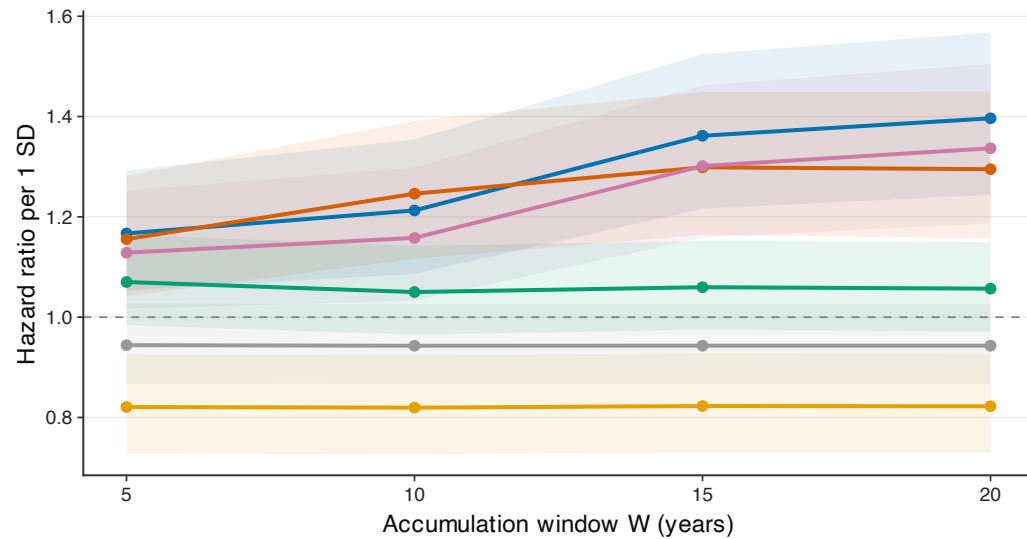


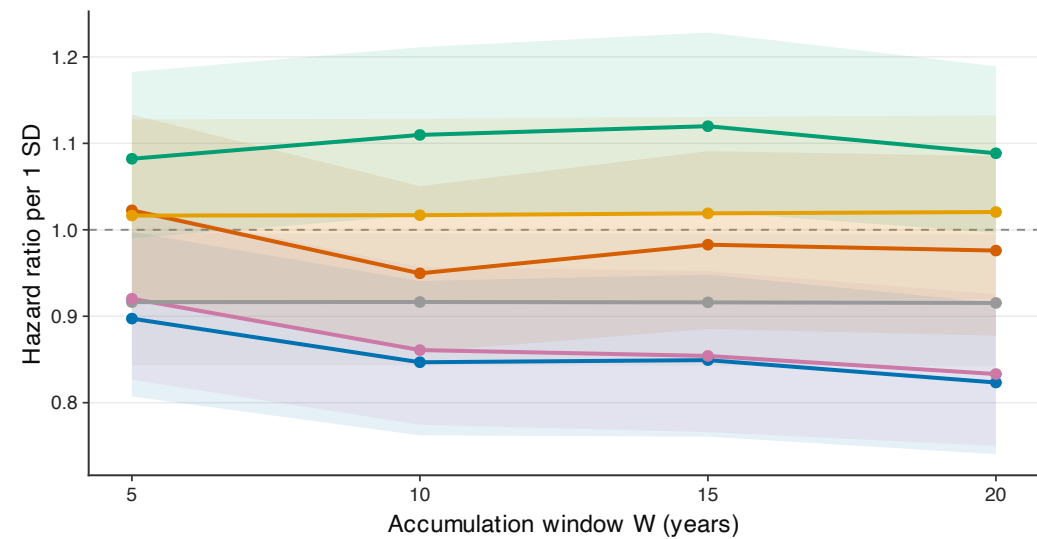
### a Every proxy carries a signal except the two niche-centre metrics

Niche proximity and niche tracking sit on 1 at every window; heat exposure is negative



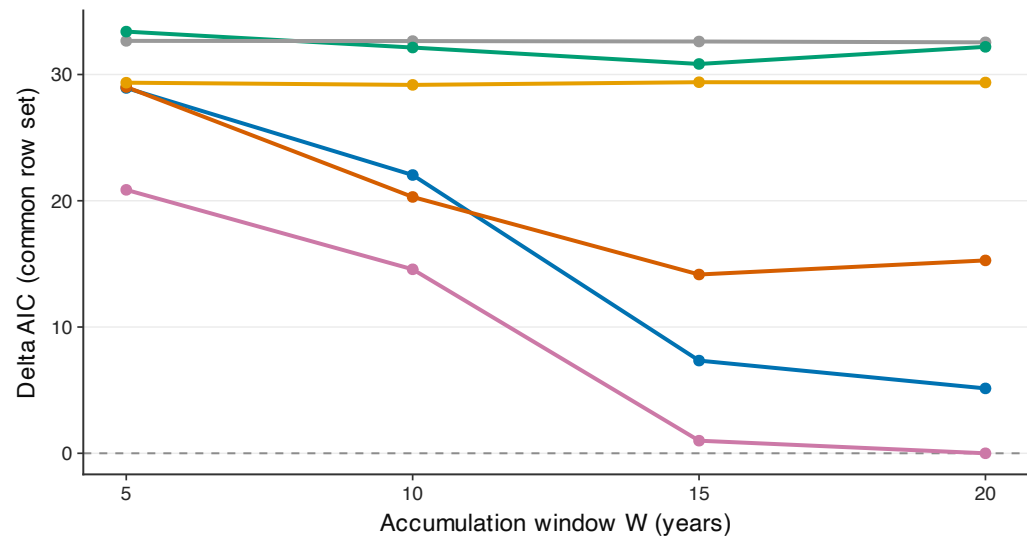
### b The interaction follows the underlying indicator, not the transformation

Both proxies built on the warmest-month maximum sit on 1 at every window. Those built on annual mean temperature depart from it - downward for the main and moderated specifications and for niche proximity, upward for niche tracking.



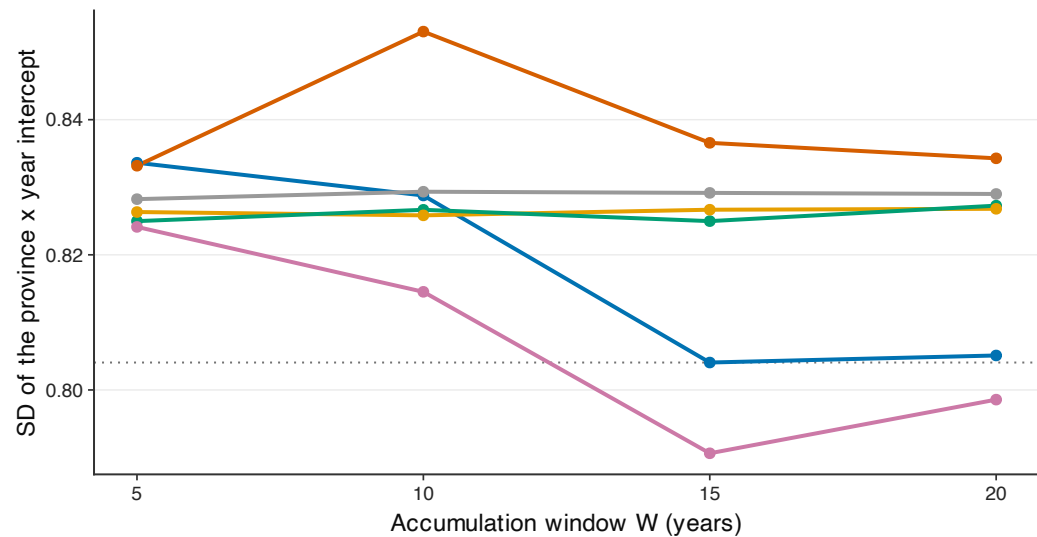
### c Heat-exposure moderation is the best specification at every window

All fits share one row set ( $n = 175,901$ ), so AIC is comparable throughout



### d A second criterion, independent of AIC

Dotted line: the main model at  $W = 15$ . Adding heat exposure absorbs province-year variance; the warmest-month proxy leaves more of it unexplained.



**Climate proxy**

- Annual mean (main)
- Annual mean + exposure
- Niche tracking
- Warmest-month max
- Heat exposure
- Niche proximity